Reliable and Large-Scale Computer System (RELACS) Research LAB

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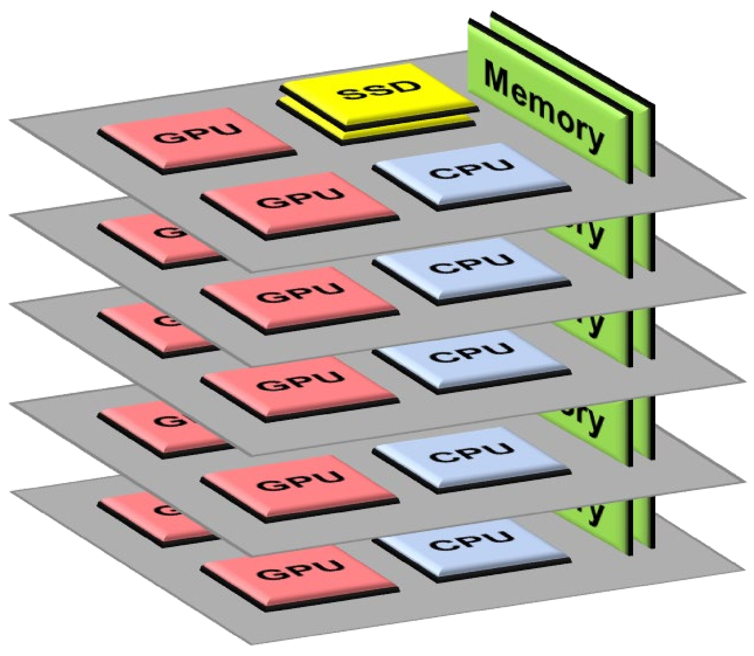
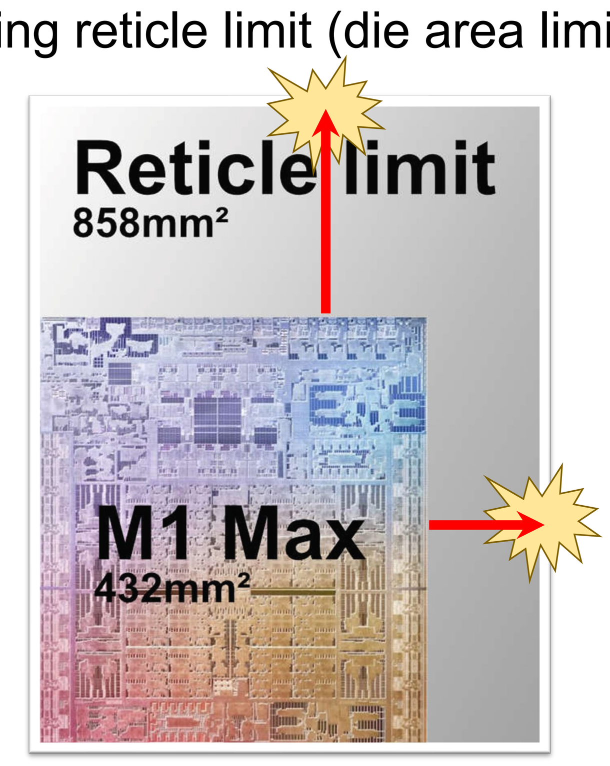
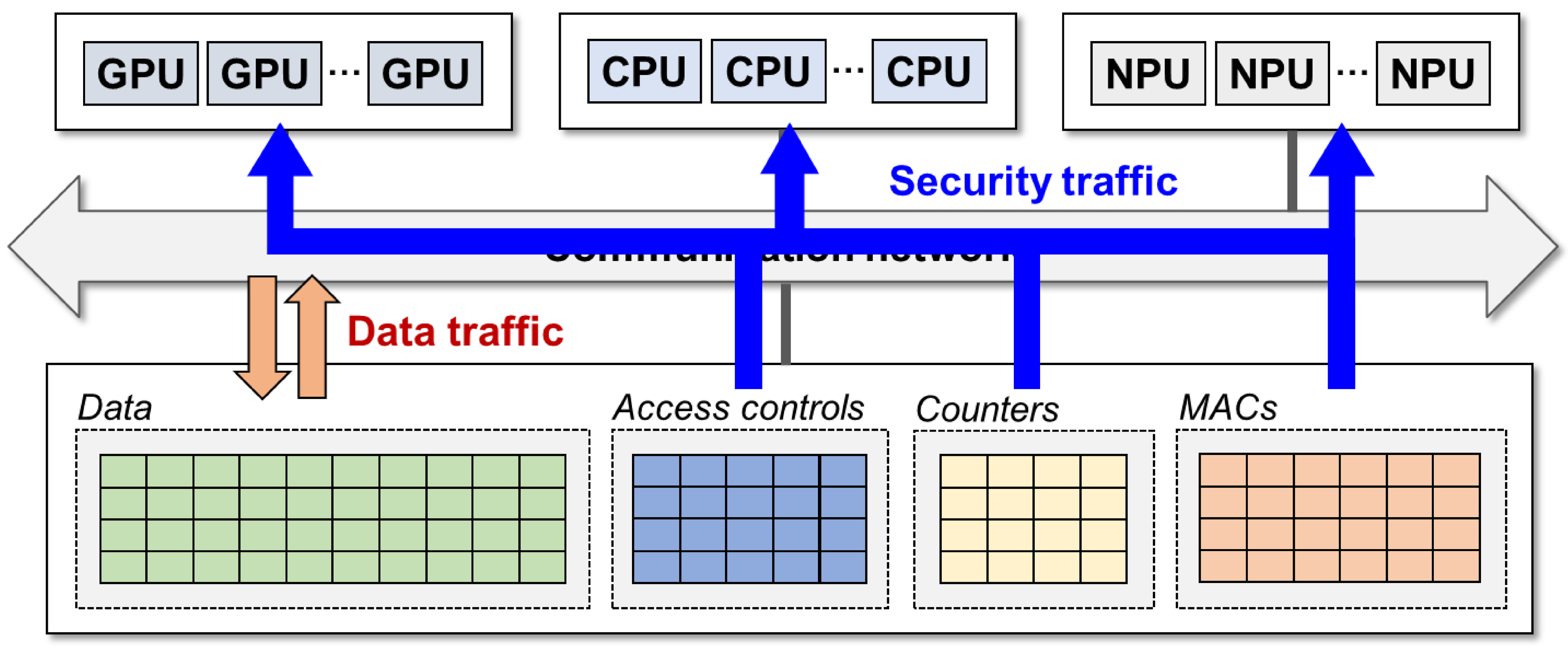
Main Research Theme: Resource Disaggregation

- Resource expansion is becoming troublesome due to form-factors…
  - System-level: Few slot counts on motherboard & saturated pin count
  - Chip-level: Transistors on a die is reaching reticle limit (die area limit)





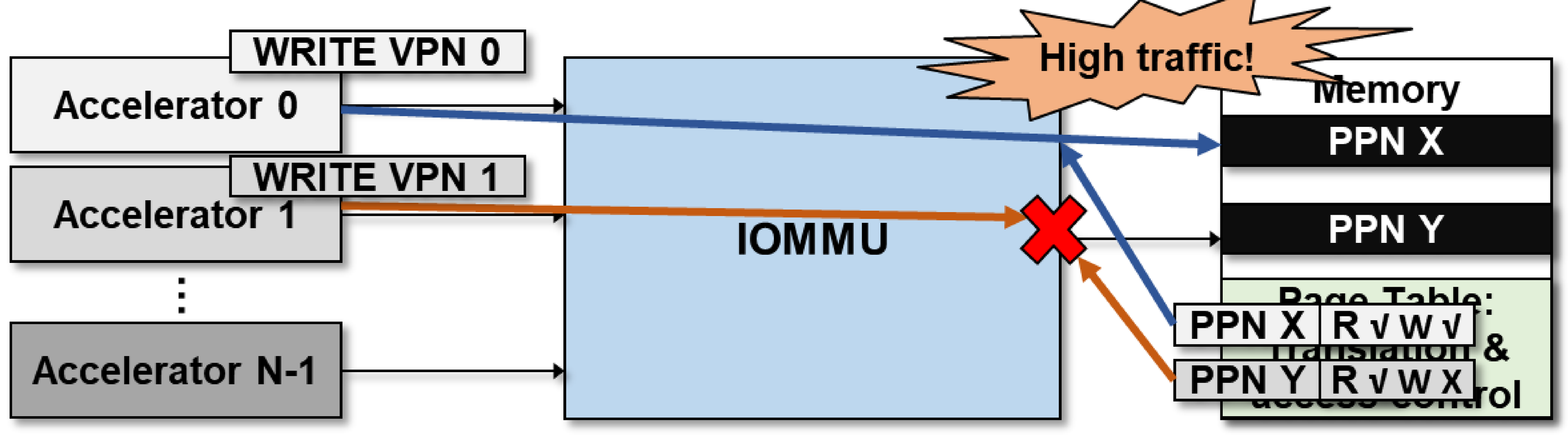
- Alternative architecture? ***Resource disaggregation***!
  - Aggregate number of similar components @ either system- or chip-level
  - Higher scalability, availability, and robustness
- ***Fundamental problem: Traffic congested on network!***



Research Topics: Security

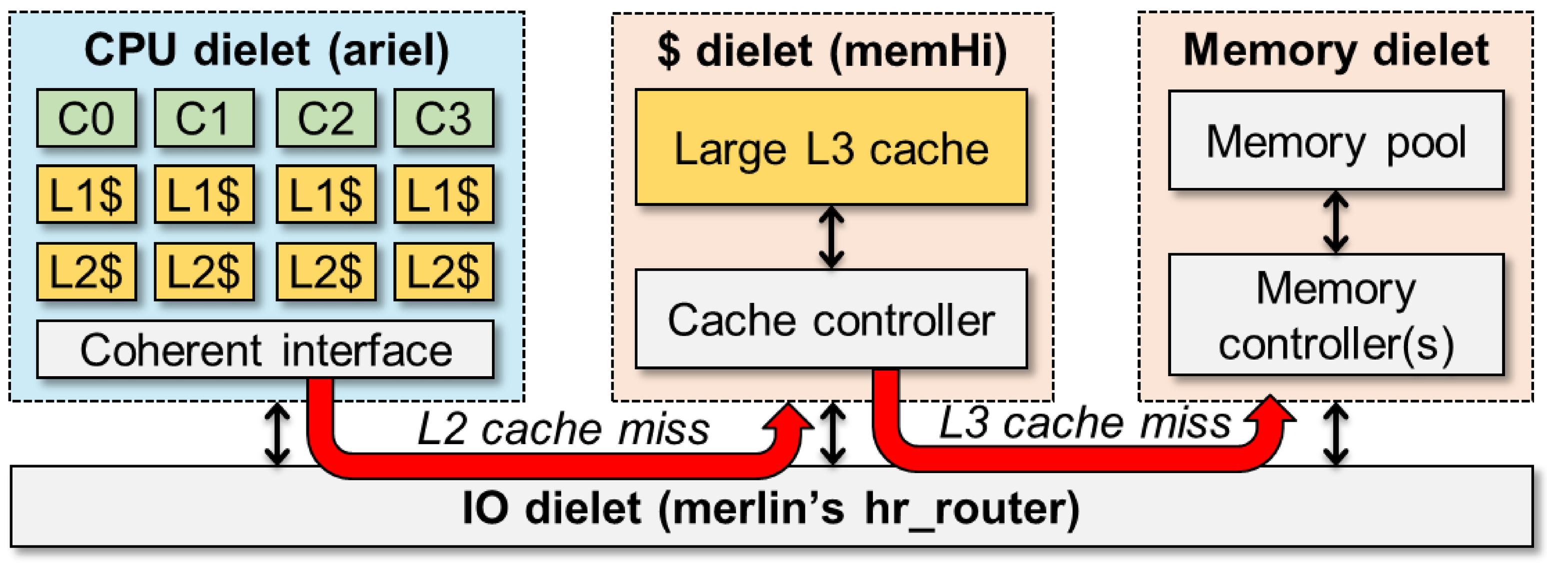
Sandboxing from Malicious Third-Party IPs

- Mission: Sandbox from malicious accelerators
  - Address translation is common in accelerator-rich systems
  - Accelerators can manipulate translation & permission information
- Mission: Sandbox from malicious IOMMU
  - IOMMU conducts address translation
  - IOMMU can also be 3rd party IP in highly disaggregated systems



Sandboxing from Malicious Memory Controller (MC)

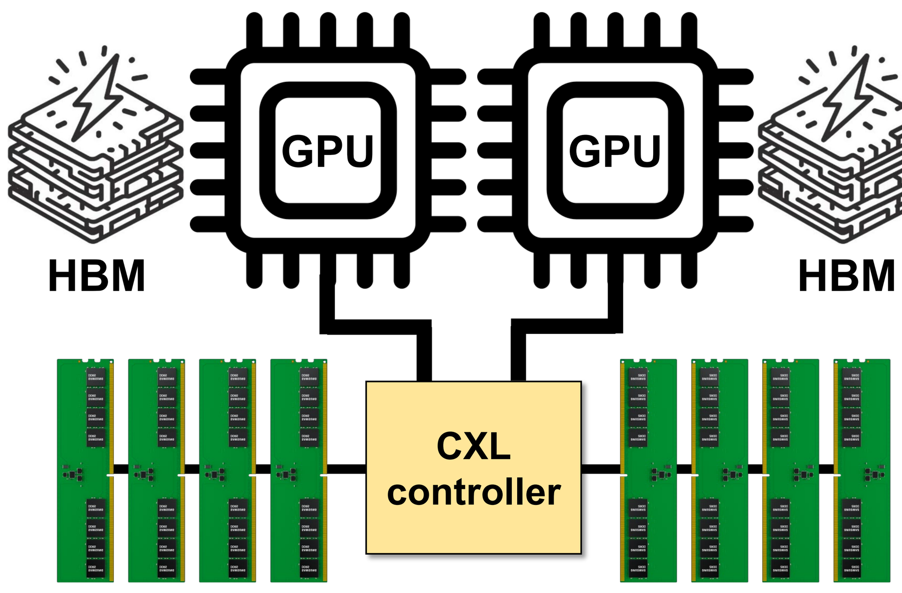
- 3rd party MC? ➔ Memory encryption engine (MEE) shouldn't be in MC
- Potential issues of decoupling MEE from MC
  - Latency exposure due to faster data fetch from L3 cache
  - Cache pollution on L3 cache for workloads having huge working set

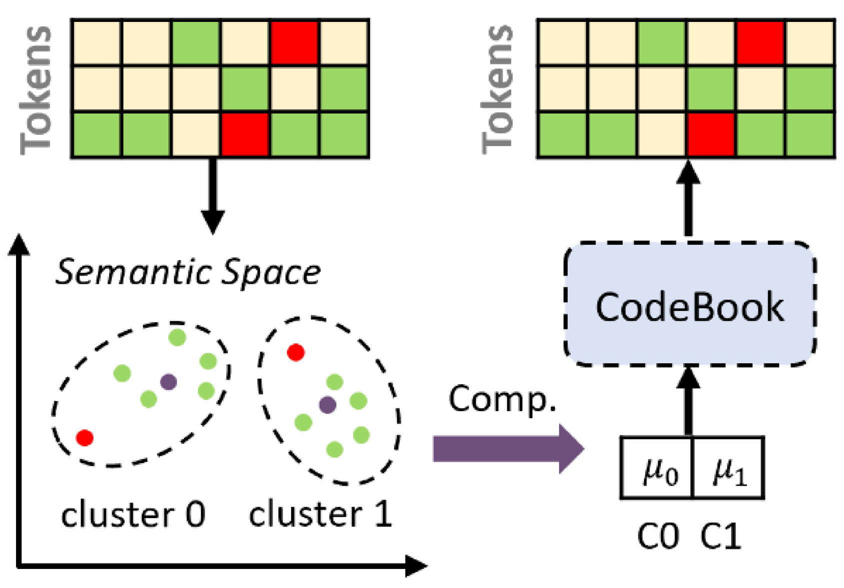


Research Topics: Compression

Lossy Compression for Large-Language Models (LLMs)

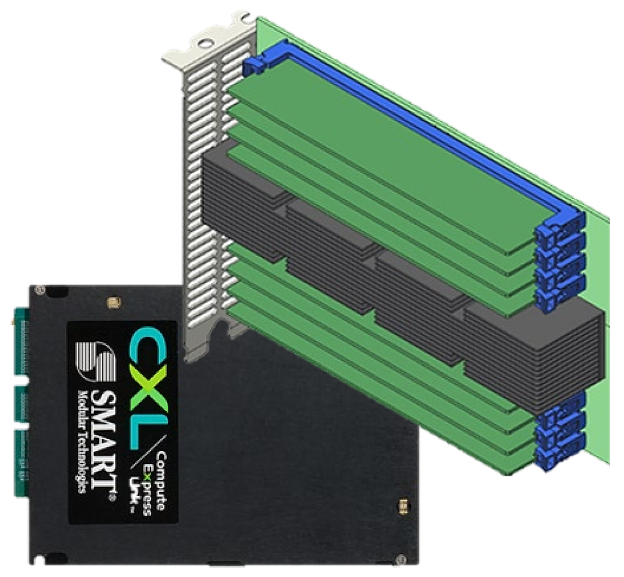
- KV cache (=context information) is a troublesome in modern LLM
  - More tokens, larger KV cache is generated → larger memory is needed
  - CXL memory is a key to accommodate large KV cache
- ***Challenge: CXL memory is much slower than HBMs***

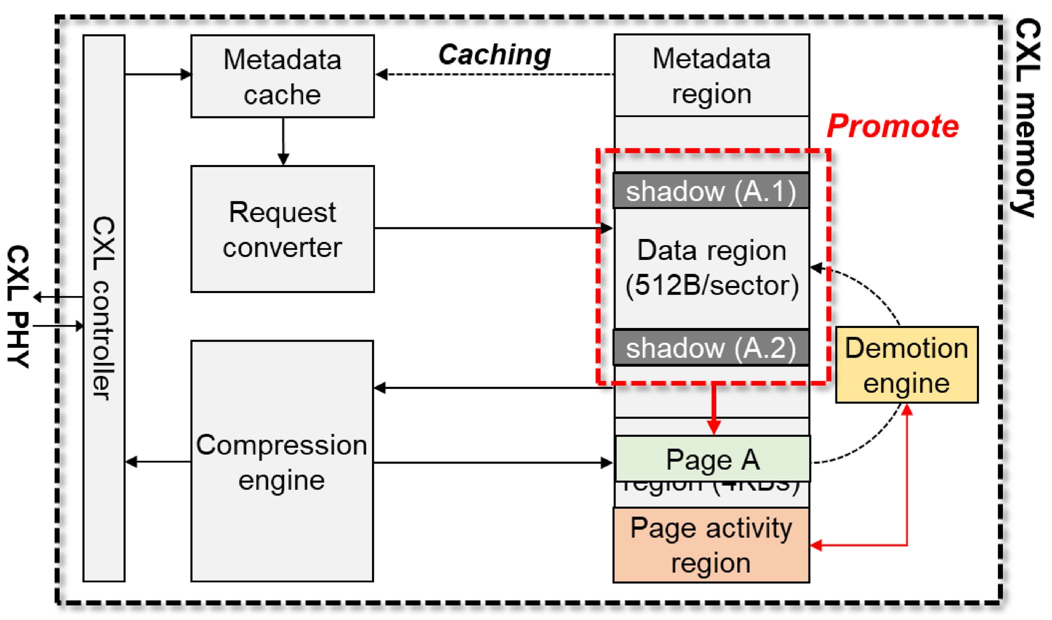




Lossless, Form-Factor-Aware Compression for CXL Memory

- CXL has been considered a strong candidate for memory disaggregation
- But its standardized form factor limit entails several ***practical challenges***
  - Small number of memory devices → limited capacity & bandwidth
  - ***Memory compression*** should be implanted inside CXL memory
- ***Challenge: Compression incurs data movements within CXL memory***





Other Information of RELACS

Benefits as a Member of RLEACS

- Close mentorship
  - You will learn how to conduct research systematically
  - Opportunities of co-working with top-tier institutions
    - IMEC, Oxford, POSTECH, SNU, SeoulTech…
  - Senior Ph.D. students can be your reliable mentor
- Attending international conferences
  - Chances to attending prestigious conference at least biyearly (But, it is more encouraged to attend as a lead author)
- Accessible to valuable infrastructure
  - High-end personal computers in lab
  - Access permission to high-end servers (512GB RAM)
  - Useful learning materials (e.g., paper list, specs, organized materials)

FAQs

- **Q1: 지원자격 요건은?**
  - 가능하면 최소 한학기 인턴쉽을 수행한 학생들 사이에서 선발
  - 오는 2학기에 지원할 경우 당장 1학기에 인턴해보는 것이 좋음
- **Q2: 연구실 진학을 위해 어떤 과목을 이수해야…?**
  - 필수: 컴퓨터구조, 디지털 논리 회로, 객체지향프로그래밍
  - 권장: 운영체제, 자료구조/알고리즘
- **Q3: 석사과정은 선발하나요?**
  - 석박통합을 우선적으로 선발할 계획
  - 분야의 학습곡선이 매우 완만하기 때문
- **Q4: 연구실에서 펀딩을(=월급) 지원해주는지?**
  - 디지스트 자체 전액 장학금 + 연구실 별도 급여 있음
- **Q5: 연구실에 출근시간이 따로 있나요?**
  - 초반에는 선배/지도교수와 close mentoring이 필요하여 출근이 필요
  - 주당 최소 55시간 투자해야 학습효과가 나옴
- **Q6: 연구실 업무는?**
  - 자신의 발전을 위한 연구에 집중 + 필요시 자신의 성과 홍보를 위한 발표

· Group Members Information

- Integrated M.S.-Ph.D. course (석박통합과정): 2 students
- Ph.D. candidate (박사과정): 2 students (from SNU and being co-advised)

**** Recruiting Integrated M.S.-Ph.D. course! 석박통합과정 모집중!***

· Achievements

- Published at 9 top-tier conference venues (MICRO, HPCA, PACT, ICS…)
* With one paper selected as Best Paper Runner-up!
- Published at 8 top-tier journals (IEEE TC, TVLSI…)
- Served as Program Committee & Reviewer for conferences & journals